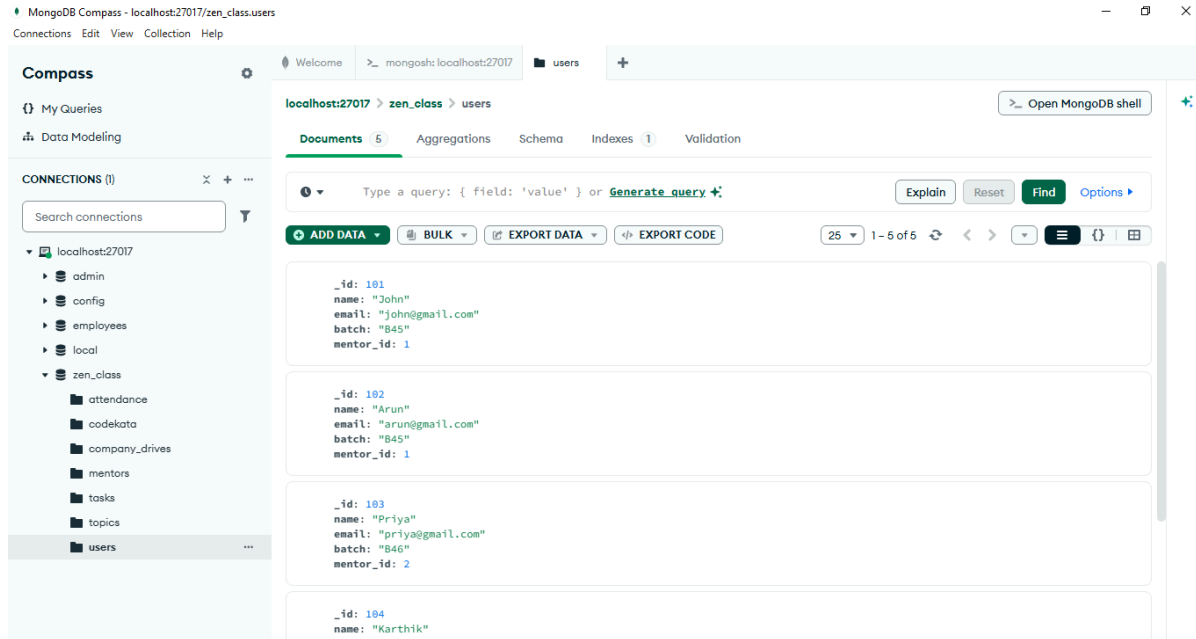


MongoDB Task

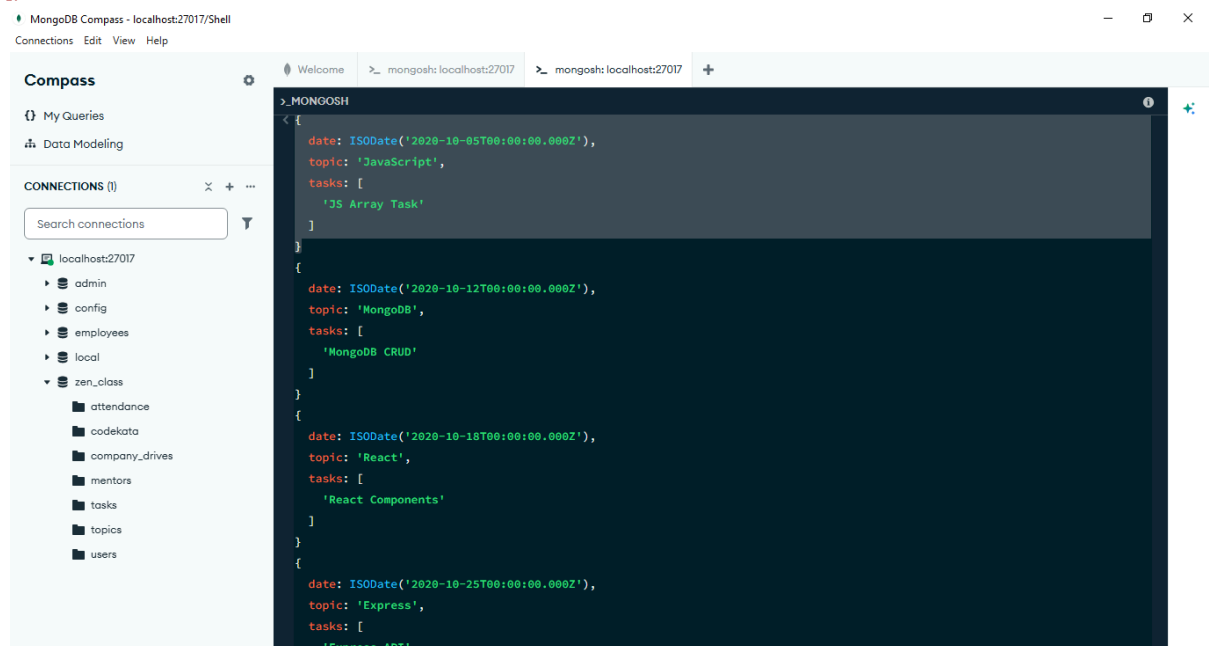


1. Find all the topics and tasks which are thought in the month of October
`db.topics.aggregate([`

```

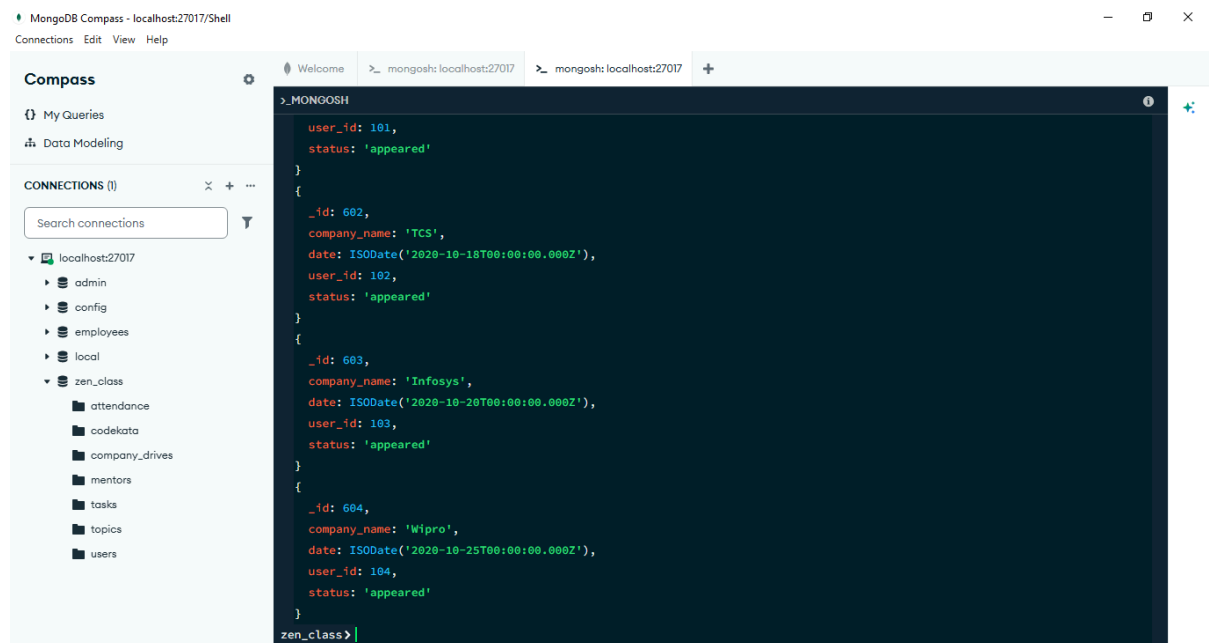
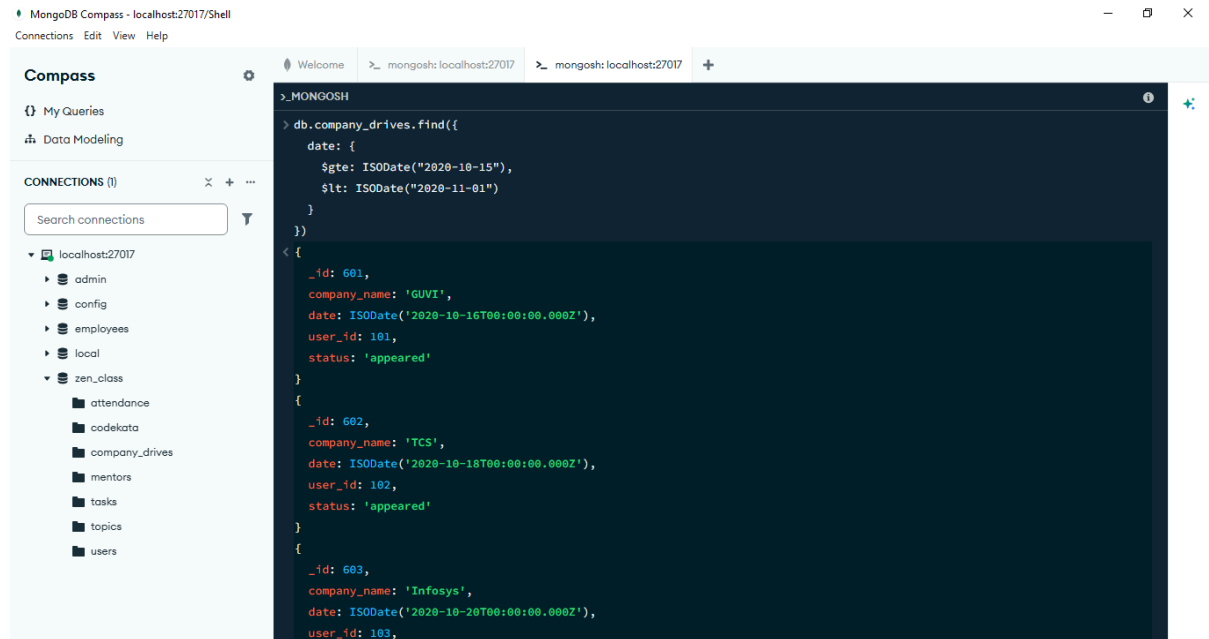
{
  $match: {
    date: {
      $gte: ISODate("2020-10-01"),
      $lt: ISODate("2020-11-01")
    }
  },
  {
    $lookup: {
      from: "tasks",
      localField: "_id",
      foreignField: "topic_id",
      as: "tasks"
    }
  },
  {
    $project: {
      _id: 0,
      topic: "$name",
      date: 1,
      tasks: "$tasks.name"
    }
  }
}
])

```



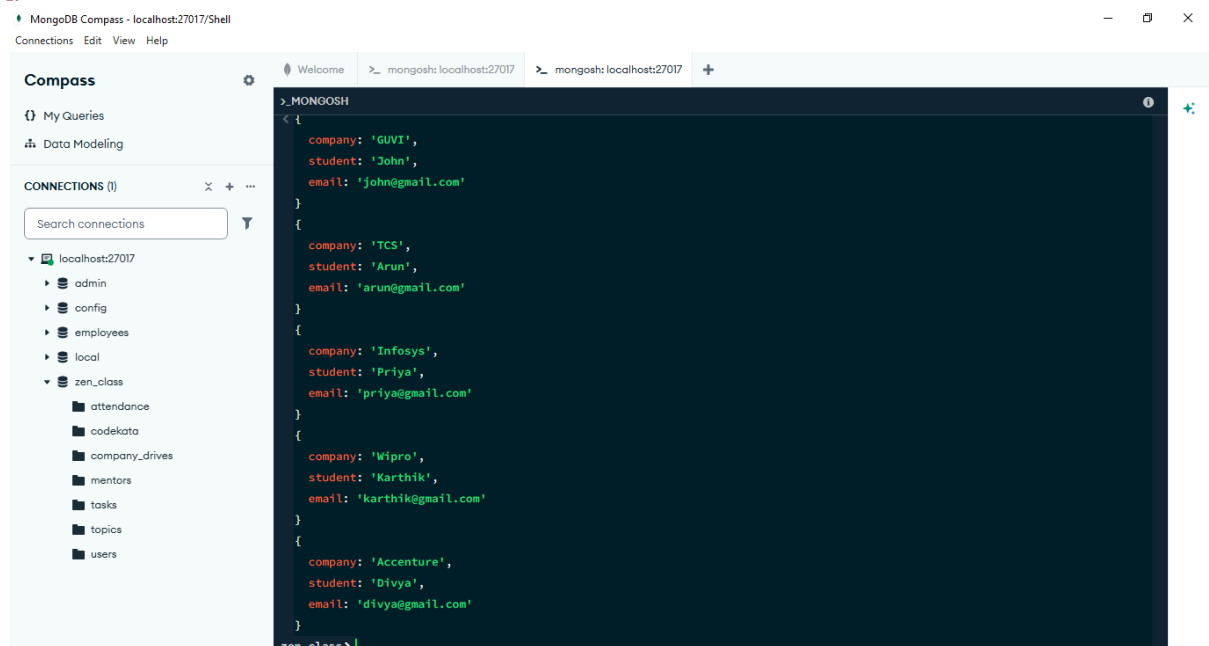
2. Find all the company drives which appeared between 15 oct-2020 and 31-oct-2020

```
db.company_drives.find({
  date: {
    $gte: ISODate("2020-10-15"),
    $lt: ISODate("2020-11-01")
  }
})
```



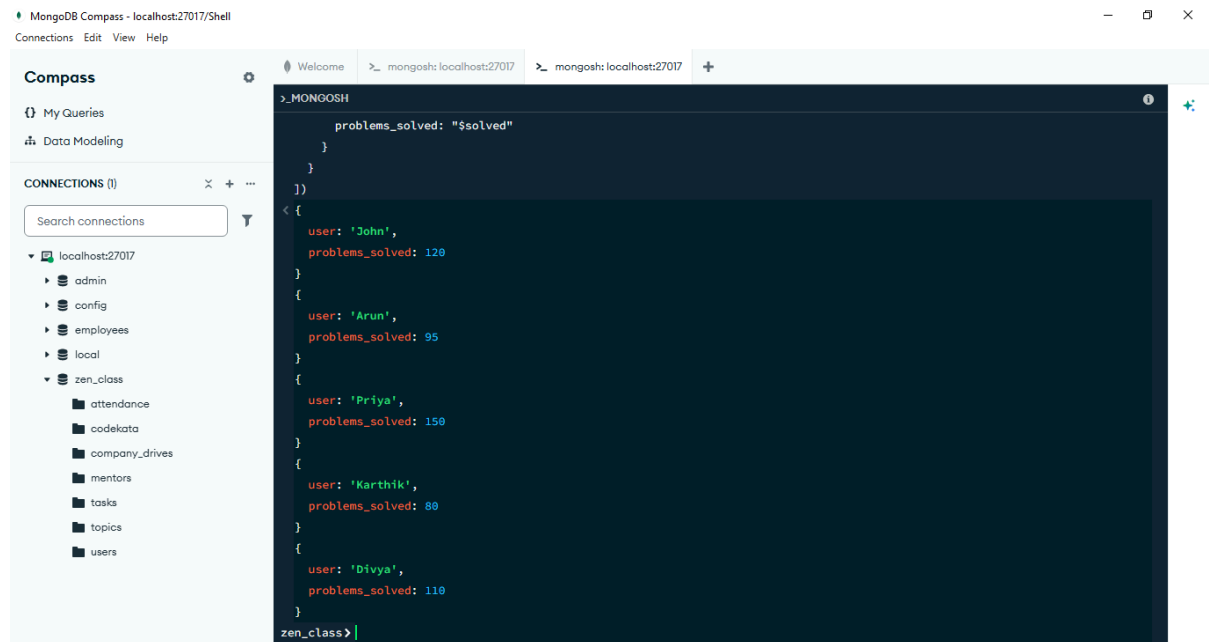
3. Find all the company drives and students who are appeared for the placement.

```
db.company_drives.aggregate([
  {
    $lookup: {
      from: "users",
      localField: "user_id",
      foreignField: "_id",
      as: "student"
    }
  },
  {
    unwind: "student"
  },
  {
    $project: {
      _id: 0,
      company: "$company_name",
      student: "$student.name",
      email: "$student.email"
    }
  }
])
```



4. Find the number of problems solved by the user in codekata.

```
db.codekata.aggregate([
  {
    $lookup: {
      from: "users",
      localField: "user_id",
      foreignField: "_id",
      as: "user"
    }
  },
  {
    unwind: "user"
  },
  {
    $project: {
      _id: 0,
      user: "$user.name",
      problems_solved: "$solved"
    }
  }
])
```



5. Find all the mentors with who has the mentee's count more than 15.

```
db.users.aggregate([
  {
    $group: {
      _id: "$mentor_id",
      mentee_count: { $sum: 1 }
    }
  },
  {
    $match: {
      mentee_count: { $gt: 2 } // more than 15
    }
  },
  {
    $lookup: {
      from: "mentors",
      localField: "_id",
      foreignField: "_id",
      as: "mentor"
    }
  },
  {
    $unwind: "$mentor"
  },
  {
    $project: {
      _id: 0,
      mentor: "$mentor.name",
      mentee_count: 1
    }
  }
])
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

employees

local

zen_class

attendance

codekata

company_drives

mentors

tasks

topics

users

Welcome

mongosh: localhost:27017

mongosh: localhost:27017

+

>_MONGOSH

```
},
{
  $lookup: {
    from: "mentors",
    localField: "_id",
    foreignField: "_id",
    as: "mentor"
  }
},
{
  $unwind: "$mentor"
},
{
  $project: {
    _id: 0,
    mentor: "$mentor.name",
    mentee_count: 1
  }
}
])
< {
  mentee_count: 5,
  mentor: 'Rajesh'
}
zen_class>
```

6. Find the number of users who are absent and task is not submitted between 15 oct-2020 and 31-oct-2020

```
db.attendance.aggregate([
  {
    $match: {
      date: {
        $gte: ISODate("2020-10-15"),
        $lt: ISODate("2020-11-01")
      },
      status: "absent"
    }
  },
  {
    $lookup: {
      from: "tasks",
      let: { uid: "$user_id" },
      pipeline: [
        {
          $match: {
            $expr: {
              $and: [
                { eq: ["user_id", "$$uid"] },
                { eq: ["status", "not submitted"] },
                { gte: ["date", ISODate("2020-10-15")] },
                { lt: ["date", ISODate("2020-11-01")] }
              ]
            }
          }
        }
      ]
    },
    as: "tasks"
  },
  {
    $match: {
      "tasks.0": { $exists: true }
    }
  },
  {
    $count: "users"
  }
])
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

employees

local

zen_class

attendance

codekata

company_drives

mentors

tasks

topics

users

Welcome

mongosh: localhost:27017

mongosh: localhost:27017

+

> MONGOSH

```
      localField: "user_id",
      foreignField: "user_id",
      as: "tasks"
    }
  },
  {
    $unwind: "$tasks"
  },
  {
    $match: {
      "tasks.status": "not submitted",
      "tasks.date": {
        $gte: ISODate("2020-10-15"),
        $lt: ISODate("2020-11-01")
      }
    }
  },
  {
    $count: "absent_and_task_not_submitted"
  }
})
< {
  absent_and_task_not_submitted: 1
}
zen_class>
```